

3 REVOLUTIONISING ANALOGUE RADIO FREQUENCY IP DEVELOPMENT IN INDIA

Silizium Circuits, an Indian startup pioneering analogue RF IP development, is reshaping India's semiconductor landscape, focusing on hard IPs and innovative protection strategies

In the dynamic landscape of the semiconductor industry, the emergence of specialised companies catering to analogue and RF design has become increasingly significant. Two friends, Dr Arun Ashok and Rijin John, with an impressive academic background and a common vision for advancing analogue radio frequency (RF) front-end intellectual property (IP) development in India, came up with an interesting semiconductor startup, Silizium Circuits.

The startup is focused on the development of analogue radio frequency IP. The company's contribution to IP development is creating hard IPs - integral components embedded within electronic devices. The strategic approach caters to companies or clients seeking to fabricate their chips but often requires external IP partnerships due to the intricacies involved. Silizium emerges as a potential partner.

Clients can allocate their resources towards core functions by offering IPs that address specific design aspects, ensuring timely product releases. The IP provisioning model, typically characterised by licensing or royalty-based agreements, not only helps clients but also generates revenue for the startup. While IPs can encompass software, the company primarily specialises in hard IP, encapsulated within a graphic data system (GDS) file format that safeguards its details.

The company utilises the Cadence digital and analogue tool-chain, complemented by various other code writing and simulation



Co-founders (L to R) Rijin John and Dr Arun Ashok

tools. However, their work encompasses more than just coding; it also involves ensuring that the written code can be transformed into a functional chip. This intricate process involves synthesis, timing analysis, and place and route procedures, all facilitated by electronic design automation (EDA) tools provided by companies like Cadence and Synopsys.

When it comes to safeguarding IP, the company adopts a comprehensive approach. They pursue domestic and international patent applications for developments with patentable potential, predominantly in the US. "In addition to this, we also do other formalities with our clients, like signing a nondisclosure agreement (NDA) document," adds Arun.

In the domain of hardware development, the company operates within the confines of foundry nodes, which essentially dictate the minimum size achievable in the manufacturing process. Being rooted in analogue and RF specialisation, the company's focus spans 65 nanometres and beyond, encompassing nodes such as 65nm, 130nm, and 180nm. A pivotal consideration in IP development is ensuring compat-

ibility with the customer's chosen technology. While their in-house products thrive in the 65nm and higher nodes, the company also accommodates the diverse technological needs of clients, extending their expertise to encompass lower technology nodes like 22nm or 28nm, as necessitated by the project at hand.

Within the Indian landscape, where most semiconductor companies are in the digital domain, Silizium Circuits is a potential partner to most of them. Comparing their IP designs with products developed internationally involves assessing various factors like power, price, performance, and area (PPA). Remarkably, Silizium's offerings outshine their foreign counterparts in these aspects, offering a superior solution. "In addition to these things, we also provide multiband support that is not present in other modules in the market globally. So, these are the two things we are scoring points where the other products are lacking," adds Arun.

"The future of the semiconductor industry in India holds immense potential. After Corona, every country is trying to think of an alternative for logistics or supply chain. India's semiconductor imports are close to the oil imports. Together with the government initiative to reduce semiconductor imports, the market is very good," says Rijin. "Just like the world has been seeing China as a powerhouse, within a few years, it will soon be India for semiconductor design, for sure," Arun adds. **EFY**

—Nidhi Agarwal